
Midterm Exam Problem 5 - Nouh Shaikh

```
clc; clear; close all;
```

```
A = [4, -3, 2;  
     2, 4, -3;  
     0, 2, 3];  
b = [16; 12; 46];
```

```
[x_gauss, L, U] = Gauss_elimination_EA(A, b);  
disp('Gauss x:'); disp(x_gauss);  
disp('L:'); disp(L);  
disp('U:'); disp(U);
```

```
y = L \ b;  
x_lu = U \ y;  
disp('LU x:'); disp(x_lu);
```

```
x0 = [0; 0; 0];  
[x_seidel] = Gauss_Seidel_EA(A, b, x0);  
disp('Gauss-Seidel x:'); disp(x_seidel);  
% So Gauss-Seidel's answer is coming out very off
```

Gauss x:

5
8
10

L:

1.0000	0	0
0.5000	1.0000	0
0	0.3636	1.0000

U:

4.0000	-3.0000	2.0000
0	5.5000	-4.0000
0	0	4.4545

LU x:

5
8
10

Iteration | Error

1	1.0000e+00
2	2.5484e+00
3	3.6803e+01
4	4.9955e+00
5	6.6853e+01
6	4.4546e+00
7	1.8592e+01
8	4.0505e+00

```
9      1.1165e+01
10     4.0496e+00
11     8.1602e+00
12     5.3311e+00
13     6.5386e+00
14     7.5260e+00
15     5.5264e+00
16     1.2128e+01
17     4.8365e+00
18     2.7784e+01
19     4.3374e+00
20     1.4641e+02
21     3.9608e+00
22     2.1601e+01
23     3.6673e+00
24     1.2097e+01
25     3.4328e+00
26     8.6070e+00
27     3.2418e+00
28     6.7992e+00
29     3.0837e+00
30     5.6966e+00
31     2.9511e+00
32     4.9558e+00
33     2.8386e+00
34     4.4255e+00
35     2.7422e+00
36     4.0282e+00
37     2.6590e+00
38     3.7205e+00
39     2.5867e+00
40     3.4757e+00
41     2.5234e+00
42     3.2770e+00
43     2.4678e+00
44     3.1130e+00
45     2.4187e+00
46     2.9758e+00
47     2.3751e+00
48     2.8596e+00
49     2.3363e+00
50     2.7603e+00
Gauss-Seidel x:
-48.8073
59.7378
-24.4918
```

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